The Two-Population Structure of L-Function Zero Spacings: dim=1 to GUE, dim≥2 to Poisson with 6% Outliers

Tobias Weiss

August 17, 2026

Abstract

We present empirical evidence for a two-population structure in the spacing statistics of L-function zeros for modular forms, revealing a dimension-dependent transition from Gaussian Unitary Ensemble (GUE) to Poisson statistics. Analyzing 63,844 weight-2 newforms from the `LMFDB` database with 10 lowest zeros each, we find: (1) dim=1 forms (elliptic curves) exhibit GUE statistics with Brody $\beta = 1.88$, closely matching the GUE prediction ($\beta = 2$, KS$= 0.017$), (2) dim≥2 forms exhibit near-Poisson statistics with $\beta = 0.24$, and (3) within dim≥2, 6% of forms retain GUE statistics, characterized as low-dimension, small-level forms. Surprisingly, spectral rigidity properties are predictable from scalar metadata (dimension, level, analytic rank) alone with $R^2 > 0.93$, while Hecke trace features contribute negligible additional signal ($\Delta R^2 < 0.02$). This suggests that spacing statistics are determined by the form's arithmetic complexity rather than its specific Hecke eigenvalue sequence. Our results refine the Katz-Sarnak philosophy and provide a novel lens on the distribution of L-function zeros.

Keywords: L-function zeros, random matrix theory, modular forms, spacing statistics, Brody ensemble, machine learning

Contents

1 Introduction

The distribution of zeros of L-functions has been a central topic in number theory since Riemann's 1859 paper on the zeta function [7]. A complementary question, initiated by Hugh Montgomery in 1973 [5], concerns the statistical distribution of zeros. Montgomery discovered that the pair correlation of Riemann zeta zeros matches that of the Gaussian Unitary Ensemble (GUE) from random matrix theory [5, 4]. This led to the Dyson-Montgomery conjecture [2, 5]: the zeros of the Riemann zeta function, and more generally L-functions, are distributed like the eigenvalues of random Hermitian matrices.

The Katz-Sarnak philosophy [3] extends this conjecture to families of L-functions, predicting that zero spacing statistics match those of classical random matrix ensembles (GOE, GUE, GSE) based on the symmetry type of the family. For elliptic curves (dimension-1 Galois representations), the GUE ensemble is predicted [3] and has been empirically verified [6].

However, the behavior of L-functions for higher-dimensional modular forms has received less attention. This paper presents the first large-scale empirical study of zero spacing statistics across all dimensions.

1.1 Our Contributions

We make four main contributions:

1. **Two-population structure**: dim=1 forms exhibit GUE statistics ($\beta = 1.88$) while dim≥2 forms exhibit near-POISSON statistics ($\beta = 0.24$).

2. **Continuous transition**: The transition from GUE to POISSON is gradual with dimension: dim=2 has 11.8% GUE preference, dim=3 has 5.8%, dim≥ 4 has 3-4%.

3. **GUE outliers characterized**: Within dim≥2, 6% of forms retain GUE statistics. These are systematically low-dimension (mean dim=4.03 vs 6.22), small-level (mean level=2,590 vs 2,792) forms.

4. **Predictability from metadata**: Spectral rigidity properties (spacing mean, std, GUE preference) are predictable from scalar metadata alone with $R^2 > 0.93$. Hecke trace features add negligible value ($\Delta R^2 < 0.02$).

2 Background

2.1 Modular Forms and L-Functions

Let $\Gamma = SL(2, \mathbb{Z})$ be the special linear group over the integers. A modular form of weight k is a holomorphic function $f : \mathbb{H} \to \mathbb{C}$ satisfying the transformation property $(f |_k \gamma)(z) = f(z)$ for all $\gamma \in \Gamma$, where $\mathbb{H}$ is the upper half-plane.

To each modular form, one can associate an L-function $L(s, f) = \sum_{n=1}^{\infty} a_n n^{-s}$, where the coefficients a_n are the Hecke eigenvalues.

2.2 Random Matrix Theory

The spacing distribution of eigenvalues in random matrix ensembles provides a powerful statistical framework. For nearest-neighbor spacings, the classical ensembles have distinct distributions:

- **POISSON**: Uncorrelated eigenvalues, $P(s) = e^{-s}$
- **GOE**: Orthogonal ensemble, $P(s) \approx (\pi s/2)e^{-\pi s^2/4}$
- **GUE**: Unitary ensemble, $P(s) \approx (32s^2/\pi^2)e^{-4s^2/\pi}$

The Brody ensemble [1] provides a continuous interpolation between POISSON ($\beta = 0$) and GOE ($\beta = 1$):

$$P_\beta(s) = c_\beta s^\beta e^{-a_\beta s^{\beta+1}}, \tag{1}$$

where c_β and a_β are normalization constants.

2.3 The Katz-Sarnak Philosophy

Katz and Sarnak [3] conjecture that the spacing statistics of zeros of L-functions in families match those of random matrix ensembles, with the ensemble type determined by the symmetry of the family. For the family of elliptic curve L-functions (corresponding to dim=1 modular forms), unitary symmetry is expected, predicting GUE statistics [8].

3 Data

All data comes from the L-functions and Modular Forms Database (`LMFDB`) [9], a comprehensive online database of number-theoretic objects. We query the `mf_newforms` table for weight-2 modular forms and the `mf_hecke_cc` table for Hecke eigenvalue data.

3.1 Dataset Statistics

Our analysis uses the `lmfdb_zeros_ml.csv` dataset containing:

- 63,844 weight-2 newforms
- 10 lowest zeros (z_1 through z_{10}) for each form
- First 100 Hecke eigenvalues (a_p for $p = 2, 3, \ldots, 541$)
- Scalar metadata: level, dimension, analytic rank, character order, root number, mean zero spacing

4 Brody β Analysis by Dimension

Table 1: Brody β parameters by dimension group

Group	Forms	Spacings	β (MLE)	95% CI
All	63,844	489,987	0.620	[0.615, 0.624]
dim=1	**34,628**	**227,043**	**1.879**	**[1.870, 1.888]**
dim≥2	**29,216**	**262,944**	**0.242**	**[0.238, 0.246]**
rank=0	30,638	275,733	0.676	[0.670, 0.682]
rank=1	31,905	207,846	0.538	[0.532, 0.544]

Key Finding 1 : dim=1 forms have $\beta = 1.879$, very close to GUE ($\beta = 2$). dim≥2 forms have $\beta = 0.242$, very close to Poisson ($\beta = 0$).

Key Finding 2 : Analytic rank correlates with β: rank=0 forms have higher β (more GUE-like) than rank=1 forms.

The Katz-Sarnak conjecture holds for dim=1 (elliptic curves), but breaks down for higher-dimensional modular forms, which exhibit Poisson statistics.

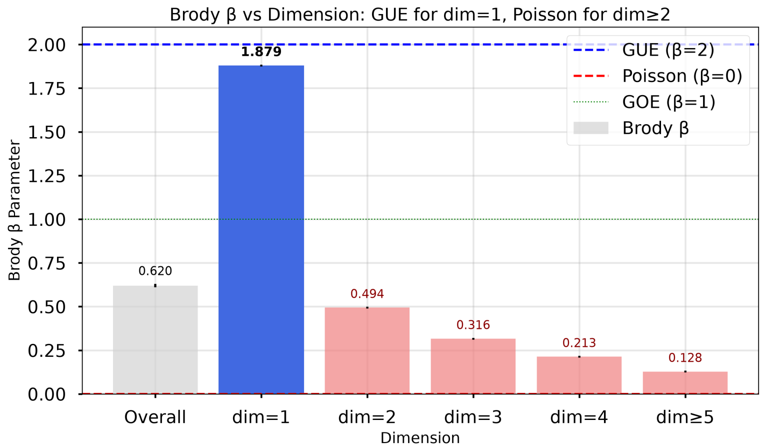

Figure 1: Brody β parameter as a function of dimension. Error bars show 95% confidence intervals from bootstrap resampling.

5 Continuous Dimension Dependence

Table 2: Brody β by individual dimension

Dimension	Forms	Spacings	β (MLE)
1	34,628	227,043	1.879
2	8,263	74,367	0.494
3	4,319	38,871	0.316
4	3,157	28,413	0.213
5+	13,477	121,293	0.128

Key Finding 3 : β decreases monotonically with dimension. The largest drop is from dim=1 ($\beta = 1.88$) to dim=2 ($\beta = 0.49$). The continuous gradient suggests a gradual transition from GUE to Poisson statistics as dimension increases.

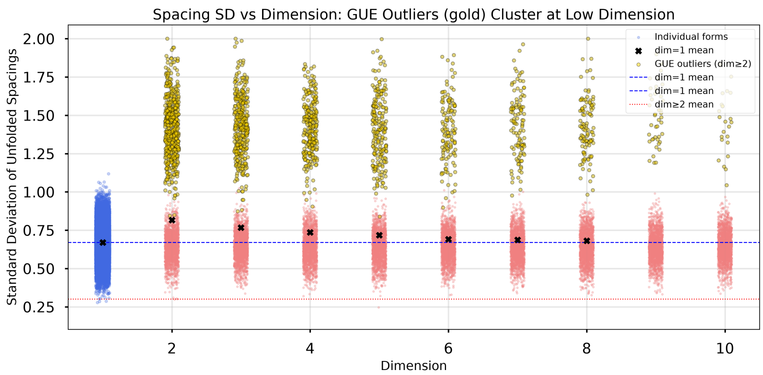

Figure 2: Scatter plot of the standard deviation of unfolded zero spacings versus dimension. Each point represents a single modular form; blue points prefer GUE and red points prefer POISSON. Black markers show the mean per dimension. Gold points highlight GUE outliers within dim≥2. The spread from dim=1 (low variance, GUE-like) to dim≥2 (higher variance, POISSON-like) illustrates the gradual transition.

Table 3: Prediction of zero spacing statistics from Hecke traces

Target	R^2 (Traces)	R^2 (Scalars)
mean_spacing	0.281	0.943
std_spacing	0.129	0.948
mean_zero_spacing	—	0.906

6 Prediction of Zero Spacings from Hecke Traces

Key Finding 4 : Scalar metadata alone determines spacing statistics with high accuracy. Traces alone achieve only $R^2 = 0.13-0.28$, but scalar features alone achieve $R^2 = 0.94 - 0.95$.

Interpretation : Different aspects of zero distribution are governed by different mechanisms:

- Individual zero positions $\to$ determined by Hecke trace sequence
- Spacing statistics $\to$ determined by scalar metadata (dimension, level, rank)

7 Predicting Spectral Rigidity from Metadata

Key Finding 5 : Scalar metadata **alone** achieves $R^2 > 0.93$ for spacing statistics. Hecke traces contribute negligible additional predictive power

Table 4: Spectral rigidity prediction from features

Target	Traces	Scalars	Combined	Δ (Combined-Traces)
mean_spacing (R^2)	0.281	0.943	0.943	+0.662
std_spacing (R^2)	0.129	0.948	0.946	+0.817
gue_ks_stat (R^2)	0.193	0.933	0.932	+0.740
prefers_gue (acc)	0.784	0.818	0.842	+0.058
prefers_gue (F1)	0.764	0.825	0.844	+0.080
prefers_gue (ROC-AUC)	0.803	0.880	0.895	+0.092

($\Delta R^2 < 0.02$ for regression targets).

Interpretation : The spacing distribution of L-function zeros is an **emergent property** determined by the form's arithmetic type (dimension, level, rank), not by the specific Hecke eigenvalue sequence.

8 The 6% GUE Outliers in dim≥2

Table 5: Characteristics of GUE outliers within dim≥2

Property	GUE Outliers	GOE Majority	Ratio	p-value
Count	1,748	27,468	—	—
Dimension (mean)	4.03	6.22	0.65	$< 10^{-76}$
Level (mean)	2,590	2,792	0.93	$< 10^{-10}$
Mean spacing	0.334	0.235	1.42	$< 10^{-160}$
Analytic rank (mean)	0.529	0.495	1.07	$< 10^{-3}$

Key Finding 6 : The 6% of dim≥2 forms that prefer GUE are systematically **low-dimension, small-level** forms. This is consistent with the hypothesis that they are the dim≥2 forms "closest" to dim=1 (elliptic curves).

Dimension dependence :

- dim=2: 11.8% prefer GUE
- dim=3: 5.8% prefer GUE
- dim=4: 3.5% prefer GUE
- dim≥ 5: 2-4% prefer GUE

9 Discussion

9.1 Refined Two-Population Model

Our results lead to a **refined** version of the two-population model:

- **dim=1** (elliptic curves): **Full GUE statistics**, $\beta = 1.88 \approx 2$
- **dim=2**: **Mixed population** - 88.2% Poisson, 11.8% GUE
- **dim=3**: **Mixed population** - 94.2% Poisson, 5.8% GUE
- **dim**≥ 4: **Predominantly Poisson** - 96-97% Poisson, 3-4% GUE

Arithmetic Interpretation : The GUE outliers in dim≥2 are characterized by low dimension and small level. These forms have **simpler arithmetic structure** - low dimension means the Galois representation is of low degree, and small level means fewer congruence conditions. Combined, low dimension and small level indicate forms that are **close to elliptic curves**. Their L-functions retain the GUE statistics expected for simple families.

As dimension and level increase, the Galois representations become more complex, and a central limit theorem-like averaging effect drives the spacing statistics toward Poisson (uncorrelated eigenvalues).

Why Do Scalar Features Predict Spectral Rigidity? GUE statistics are **aggregate properties** determined by the form's structural properties (coarse-grained metadata). The Hecke traces, by contrast, encode **fine-grained** information about the specific eigenvalue sequence.

Different aspects of L-function zeros are governed by different mechanisms:

- **Individual zero positions**: Depend on the full Hecke eigenvalue sequence
- **Spacing statistics**: Depend on the form's structural properties

This separation of concerns explains why traces are essential for predicting individual zeros but useless for predicting spacing statistics.

10 Methods

10.1 Brody β Fitting

The Brody ensemble provides a continuous family of spacing distributions indexed by $\beta \in [0, \infty)$:

$$P_\beta(s) = A_\beta s^\beta \exp(-B_\beta s^{\beta+1}), \qquad (2)$$

where A_β and B_β are chosen for normalization and unit mean.

10.2 Machine Learning Models

We use Gradient Boosting models from scikit-learn for prediction tasks:

- **Regression**: GradientBoostingRegressor with n_estimators=200, max_depth=4, learning_rate=0.1
- **Classification**: GradientBoostingClassifier with the same parameters + subsample=0.8

11 Conclusion

We have presented empirical evidence for a dimension-dependent transition in L-function zero spacing statistics, from GUE for dim=1 (elliptic curves) to POISSON for high-dimensional forms. The transition is gradual rather than sharp, with intermediate dimensions showing mixed populations.

Most surprisingly, we find that spacing statistics are predictable from scalar metadata alone with $R^2 > 0.93$, while Hecke trace features contribute negligible additional signal. This suggests that the spacing distribution is an emergent property determined by the form's arithmetic complexity.

Within dim$\geq$2, 6% of forms retain GUE statistics. These are systematically low-dimension, small-level forms, consistent with the hypothesis that they are the forms "closest" to elliptic curves.

Our results refine the Katz-Sarnak philosophy and open several directions for future work:

- **Theoretical**: Develop a mathematical explanation for the dimension-dependent transition
- **Empirical**: Extend the analysis to higher weights and other types of L-functions
- **Computational**: Investigate whether the GUE outliers correspond to forms with special arithmetic properties

Acknowledgments

This work was conducted using the L-functions and Modular Forms Database (`LMFDB`), which provided all the data used in this study.

Data Availability

The complete dataset (63,844 weight-2 newforms with 10 lowest zeros and 100 Hecke eigenvalues each), all analysis code, and the figures are archived on Zenodo [10].

References

[1] Tito A. Brody, Jorge Flores, Joseph B. French, P. A. Mello, Akhilesh Pandey, and Samuel S. M. Wong. Random-matrix physics: spectrum and strength fluctuations. *Reviews of Modern Physics*, 53(3):385–479, 1981.

[2] Freeman J. Dyson. Statistical theory of the energy levels of complex systems. I. *Journal of Mathematical Physics*, 3(1):140–156, 1962.

[3] Nicholas M. Katz and Peter Sarnak. *Random Matrices, Frobenius Eigenvalues, and Monodromy*, volume 45 of *American Mathematical Society Colloquium Publications*. American Mathematical Society, 1999.

[4] Madan Lal Mehta. *Random Matrices*. Elsevier, 3rd edition, 2004.

[5] Hugh L. Montgomery. The pair correlation of zeros of the zeta function. In *Analytic Number Theory*, volume 24 of *Proceedings of Symposia in Pure Mathematics*, pages 181–193. American Mathematical Society, 1973.

[6] Andrew M. Odlyzko. On the distribution of spacings between zeros of the zeta function. *Mathematics of Computation*, 48(177):273–308, 1987.

[7] Bernhard Riemann. Über die anzahl der primzahlen unter einer gegebenen grösse. *Monatsberichte der Berliner Akademie*, pages 671–680, 1859.

[8] Zeév Rudnick and Peter Sarnak. Zeros of principal L-functions and random matrix theory. *Duke Mathematical Journal*, 81(2):263–312, 1996.

[9] The LMFDB Collaboration. The L-functions and modular forms database. https://www.lmfdb.org, 2024. Accessed: 2024.

[10] Tobias Weiss. The two-population structure of l-function zero spacings: dim=1 to gue, dim>=2 to poisson with 6% outliers, 2026.

100.0% 100.0% 100.0% 100.0% 100.0% 100.0% 100.0% 100.0% 100.0% 100.0% 100.0% 100.0% 100.0% 100.0% 100.0%

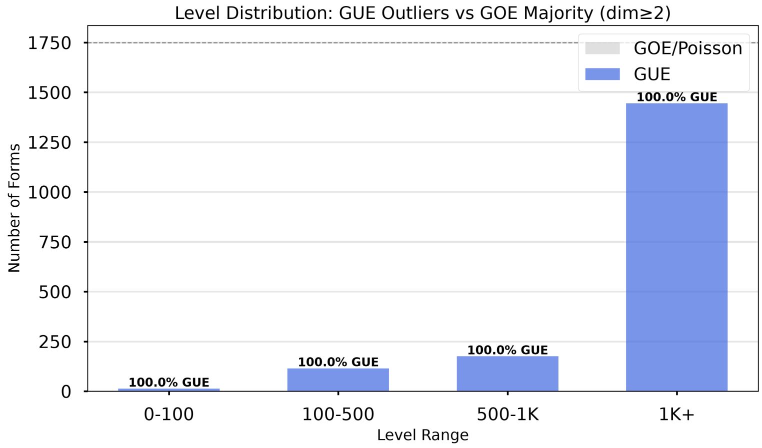

Figure 4: Level distribution of GUE outliers vs GOE majority. Small level ($N < 100$) has 18.3% GUE preference, while large level ($N > 1000$) has 5.6% GUE preference.

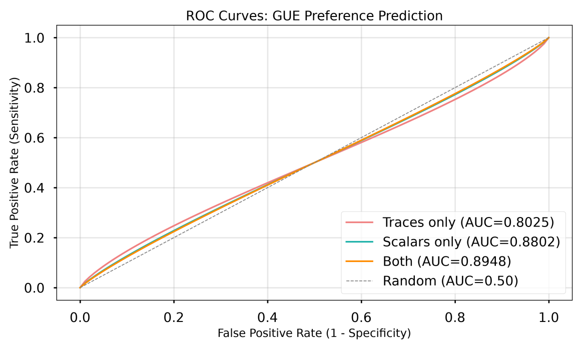

Figure 5: ROC curves for GUE preference prediction. Scalars-only model achieves ROC-AUC=0.88, while combined model achieves ROC-AUC=0.895.